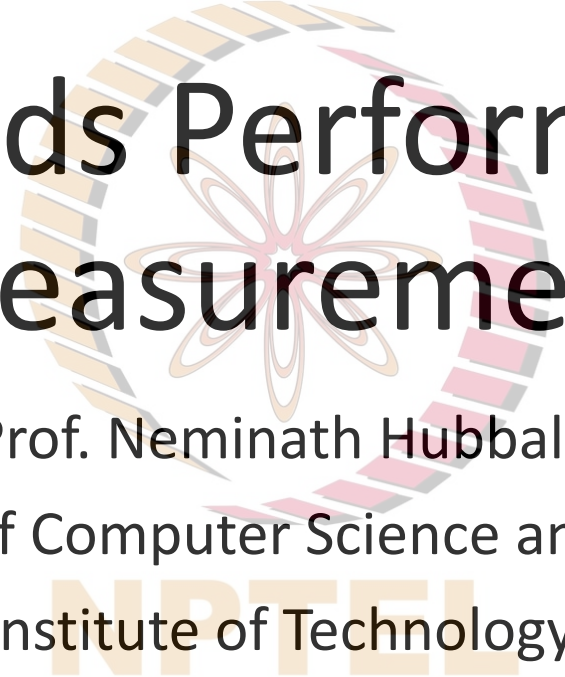


Towards Performance Measurement

A large, semi-transparent watermark of the NPTEL logo is centered in the background. It features a circular emblem with a stylized flower or gear design in the center, surrounded by a ring of colored segments. Below the emblem, the word "NPTEL" is written in large, bold, orange capital letters.

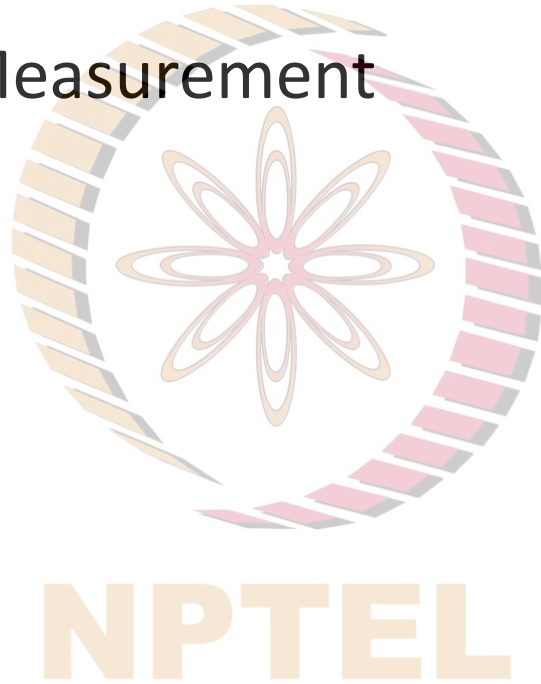
Prof. Neminath Hubballi

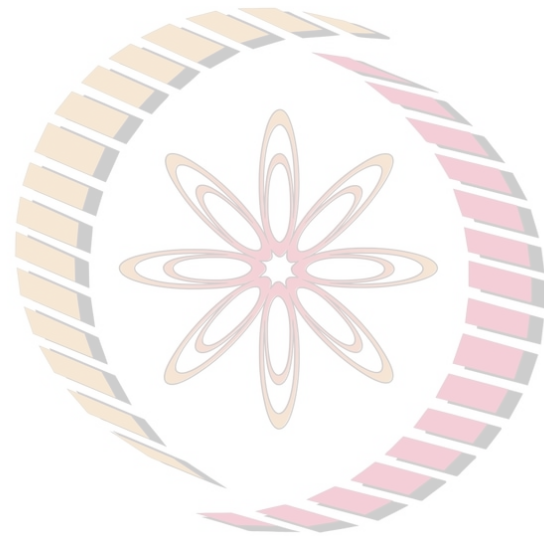
Department of Computer Science and Engineering

Indian Institute of Technology Indore

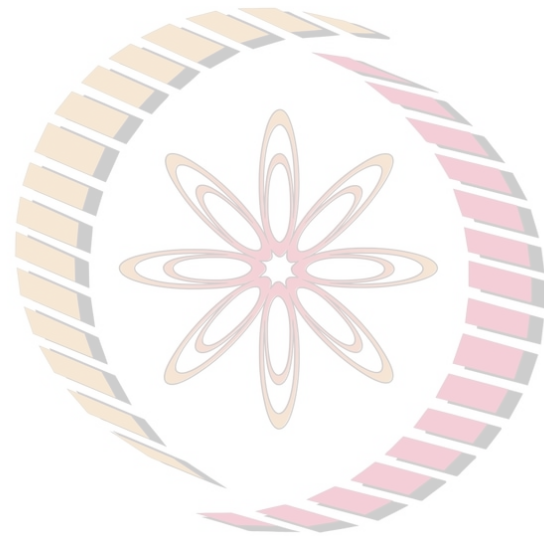
Agenda

- ☐ Towards Performance Measurement
 - ☐ Transmission delay
- ☐ Circuit Switching
- ☐ Packet Switching



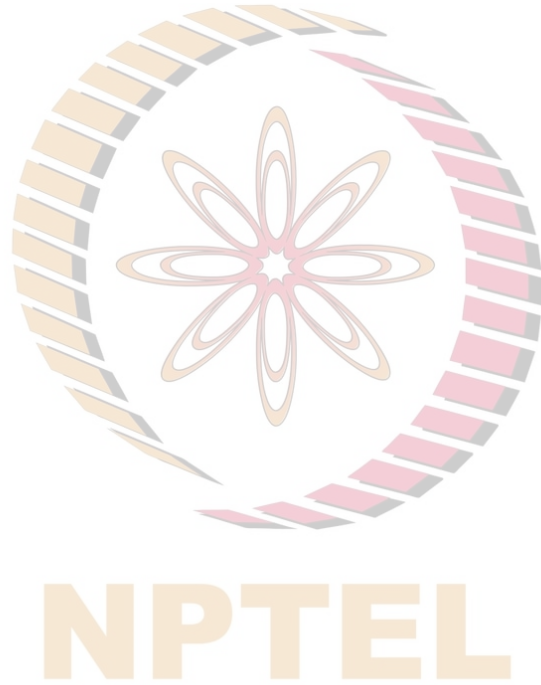


NPTEL

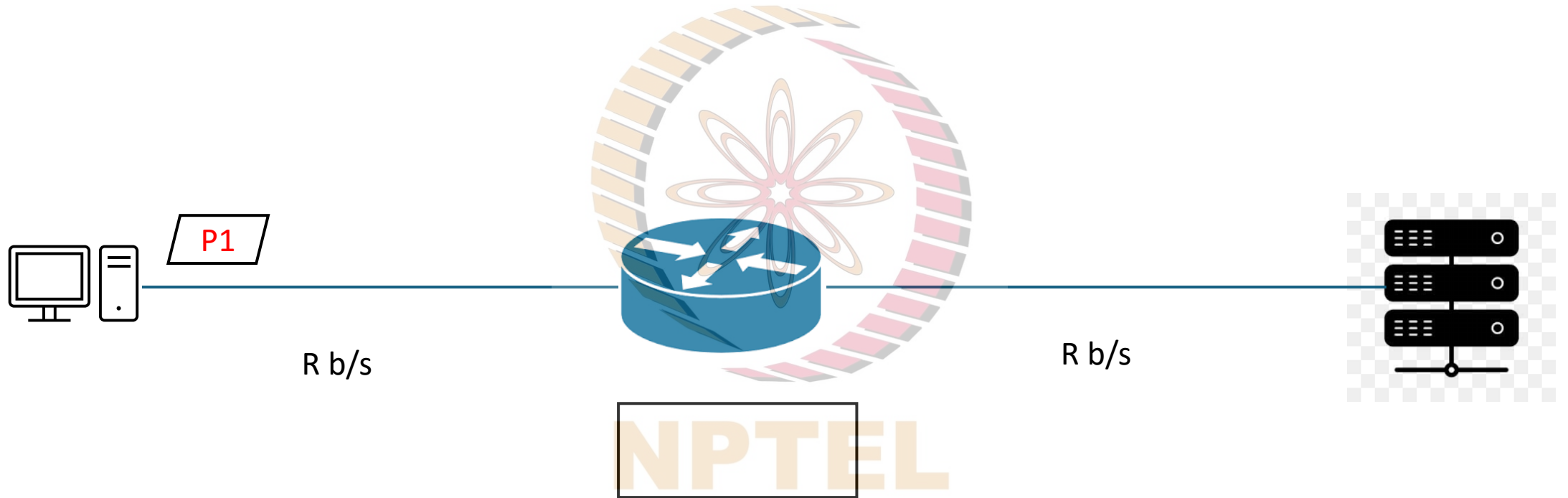


NPTEL

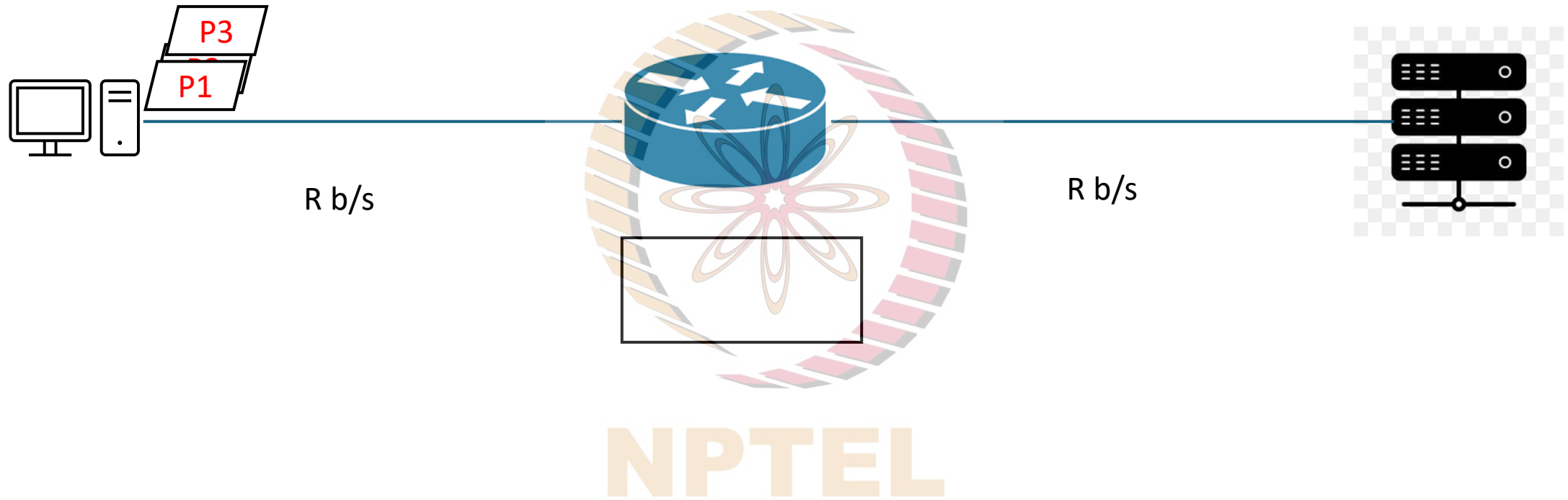
Transmission Delay



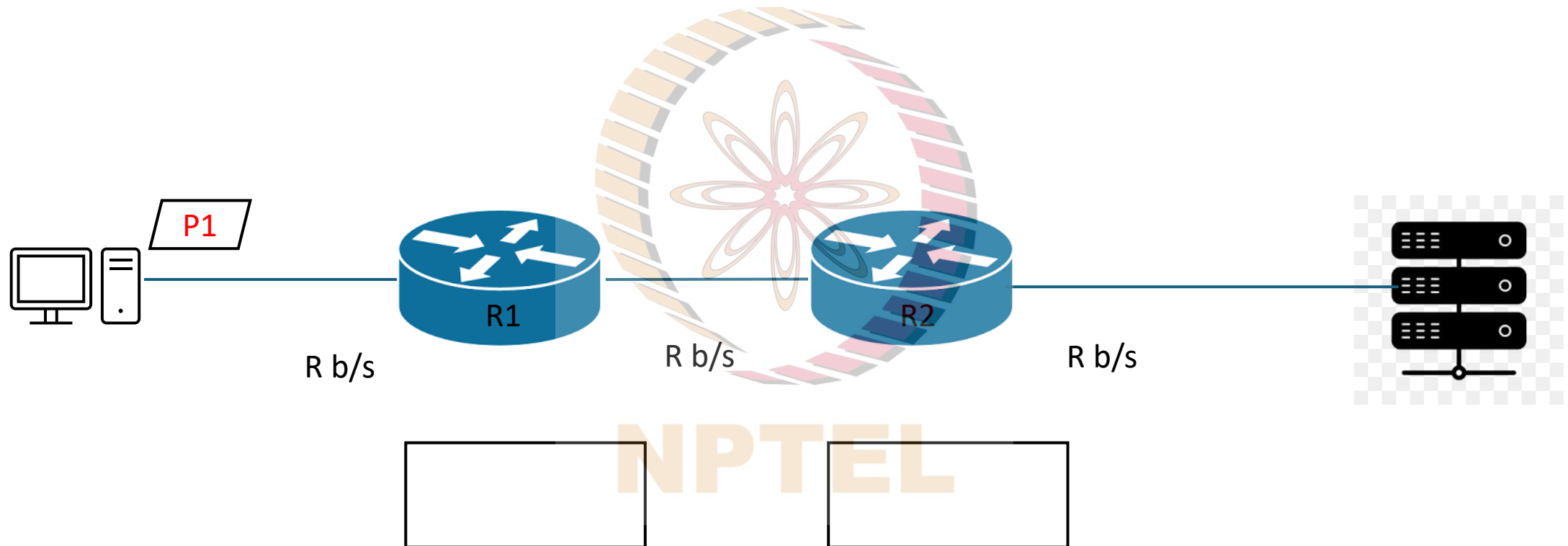
Transmission Delay: One Packet One Intermediate Router



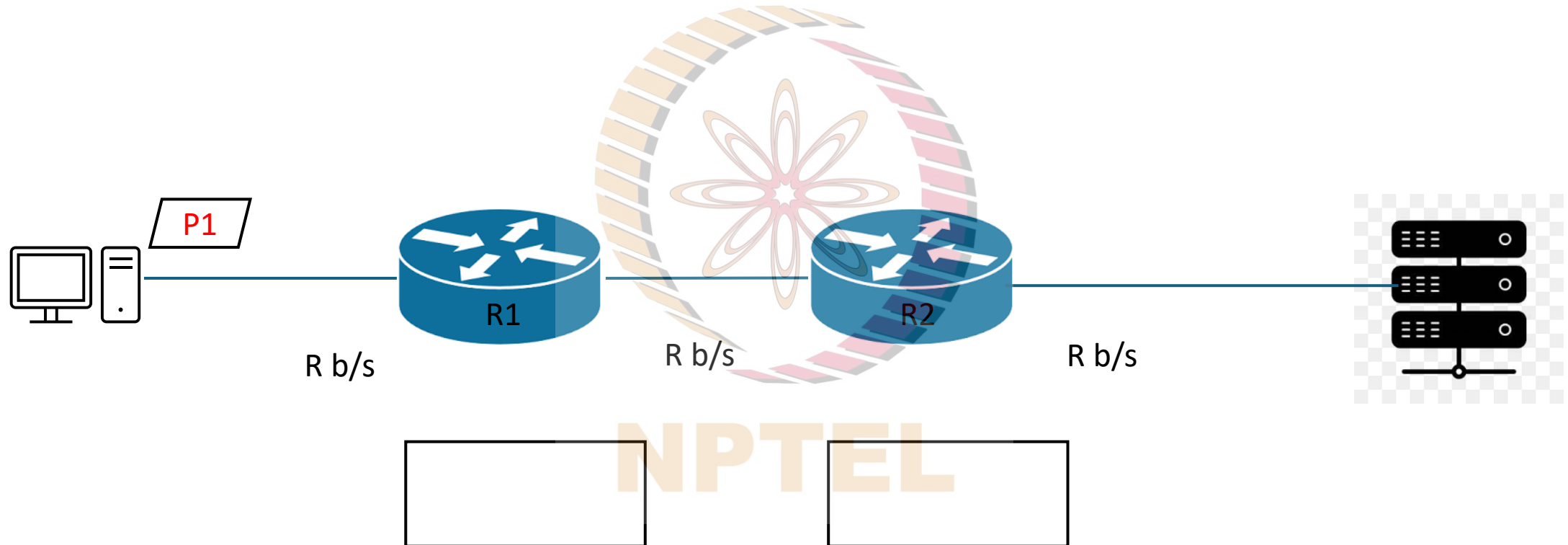
Transmission Delay: Multiple Packets and One Intermediate Router



Transmission Delay: One Packet Multiple Hops

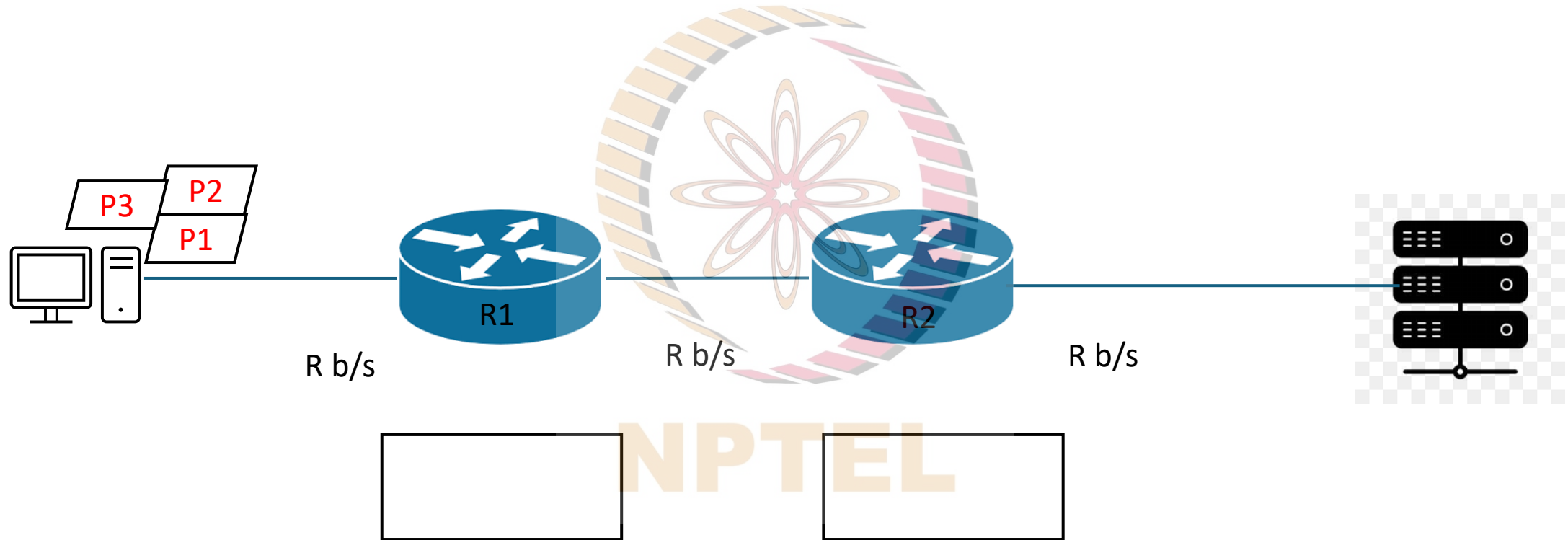


Transmission Delay: One Packet Multiple Hops

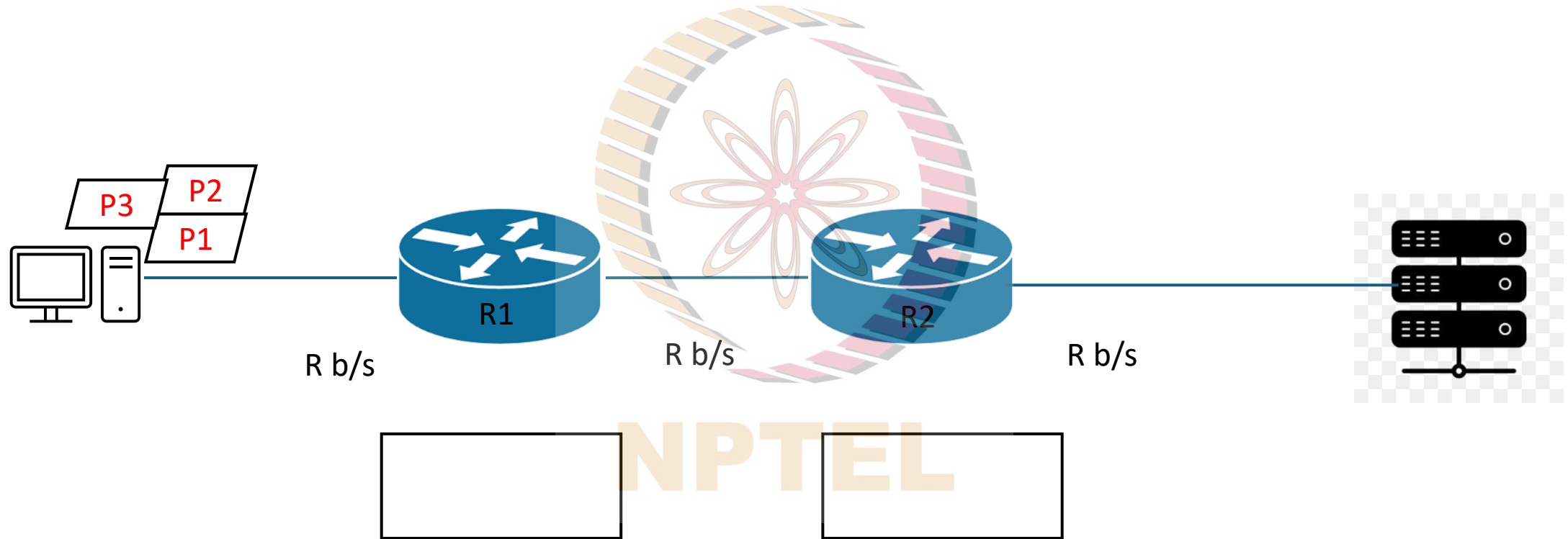


$$d_{trans_total} = N \times \frac{L}{R}$$

Transmission Delay: Multiple Packets and Multiple Hops



Transmission Delay: Multiple Packets and Multiple Hops



$$d_{total_trans} = (N + P - 1) \times \frac{L}{R}$$

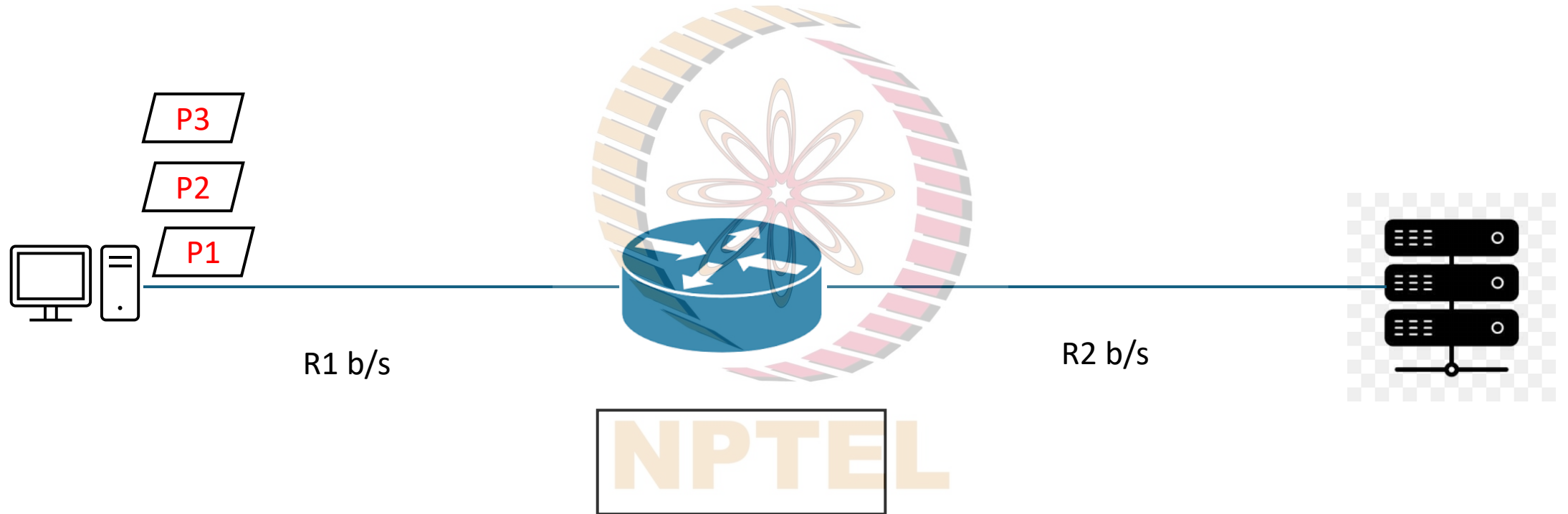
Transmission Delay: Multiple Packets and Multiple Hops

□ If you have **3 hops** ($N=3$), **10 packets** ($P=10$), and each packet takes **1ms** to transmit ($L/R=1\text{ms}$):

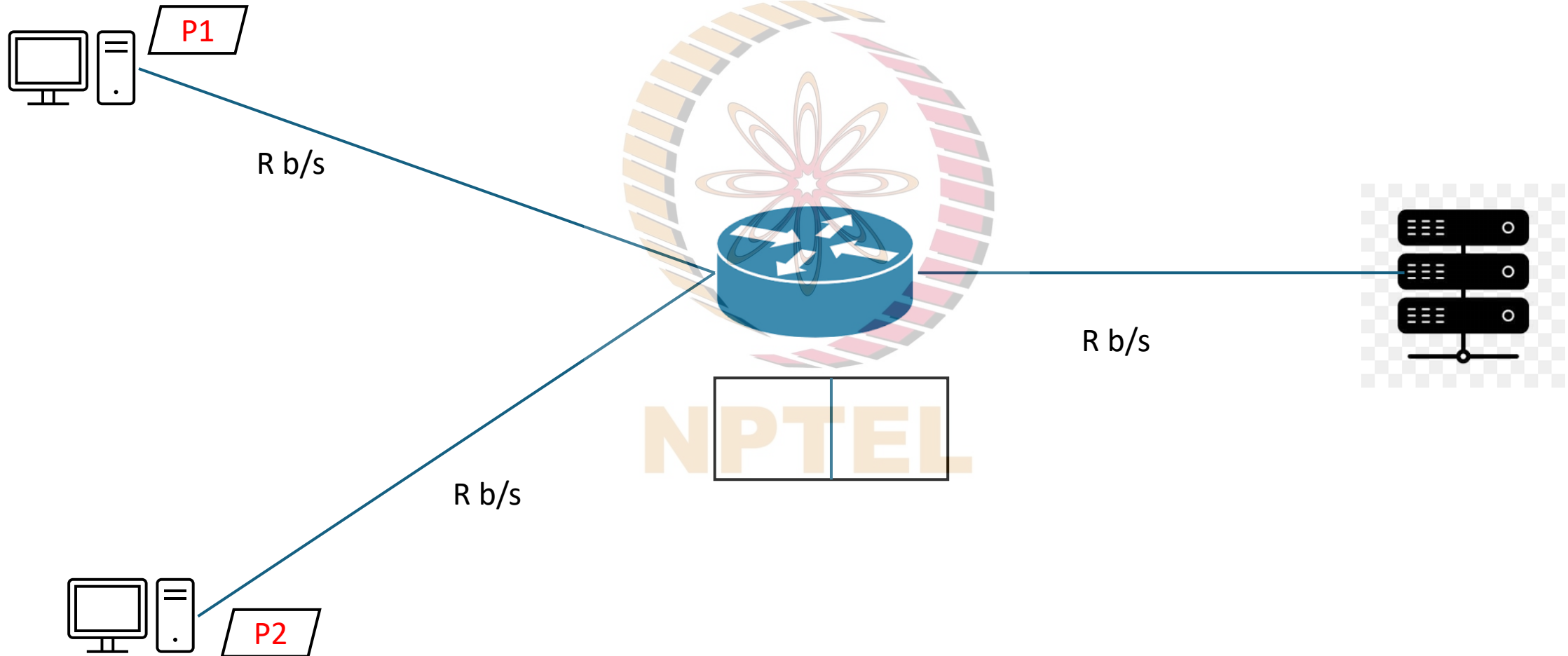
$$d_{total_trans} = (3 + 10 - 1) \times 1\text{ms} = 12\text{ms}$$

NPTEL

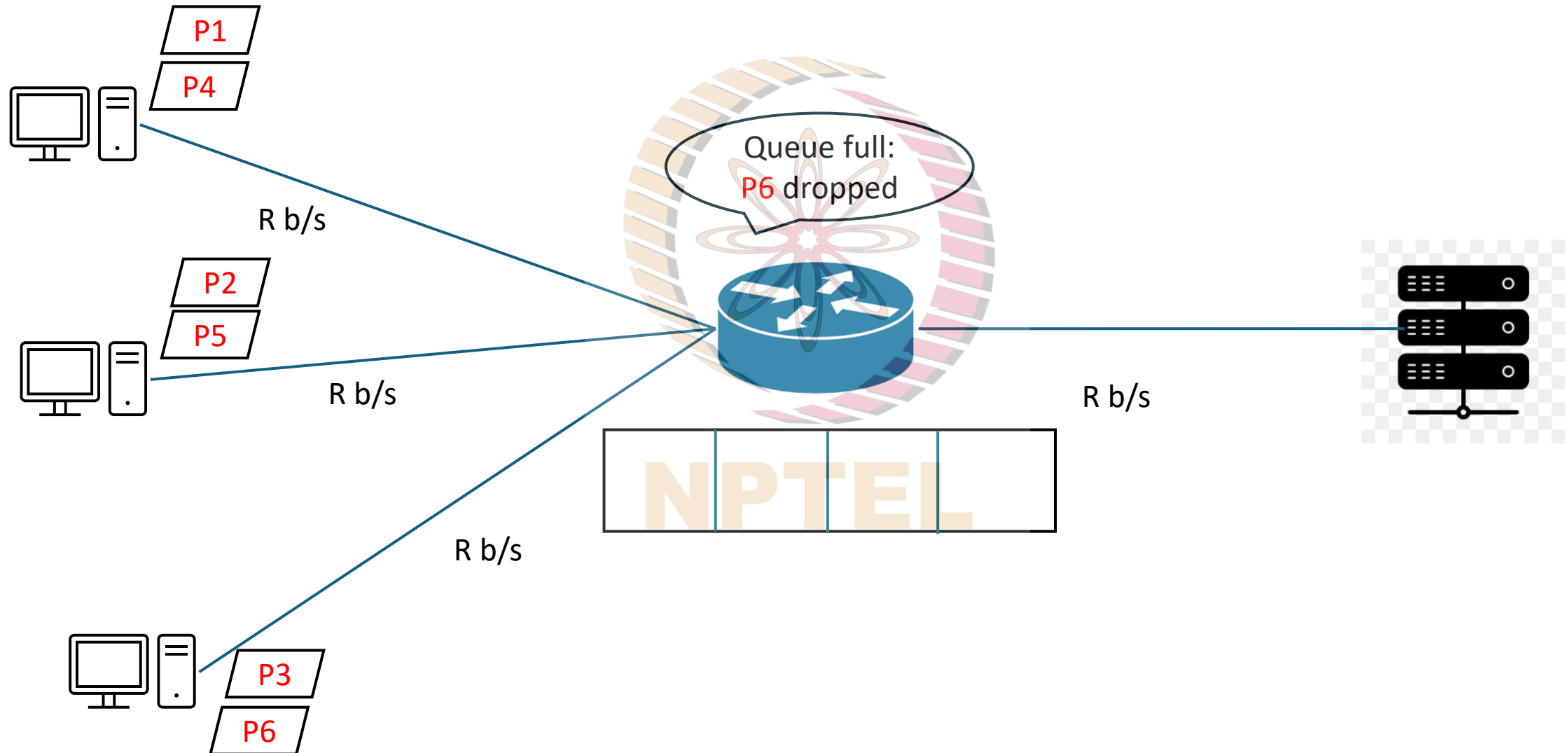
Disparity in Transmission Rates



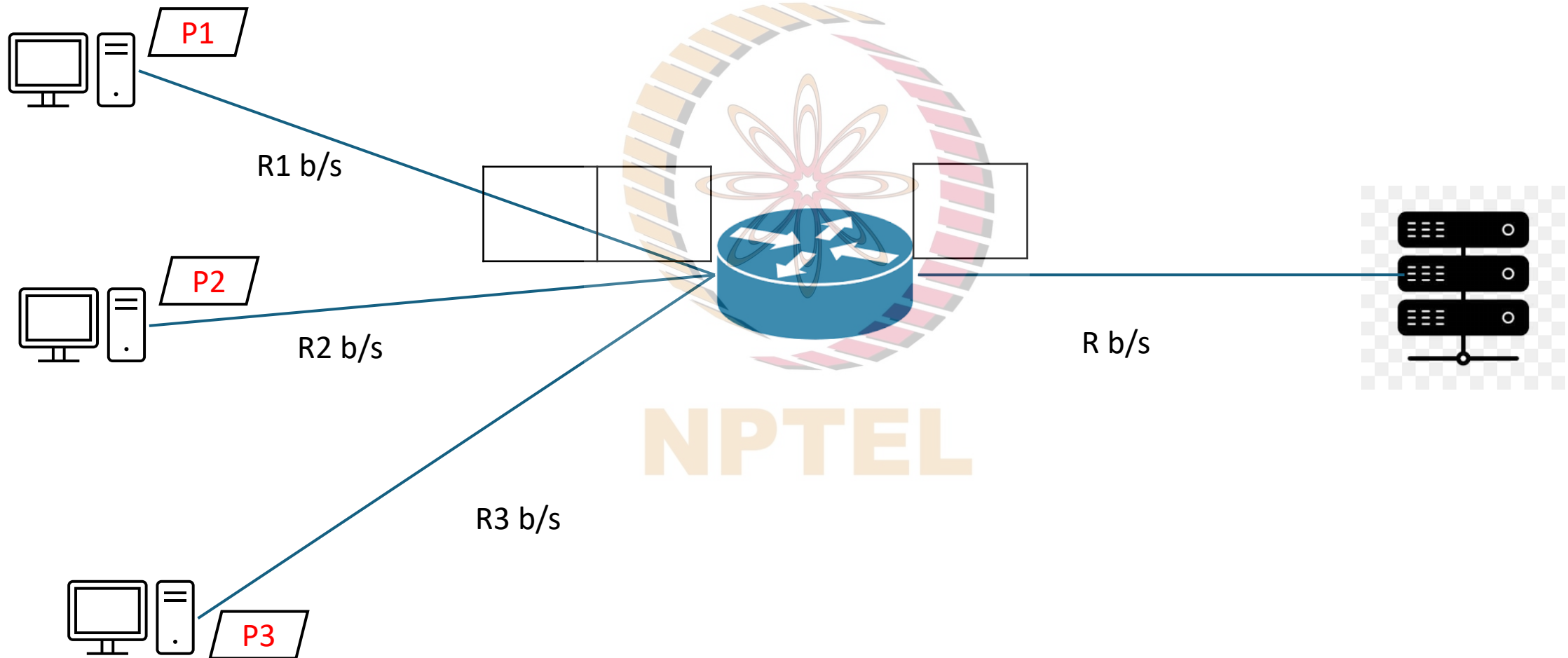
Multiplexing



Multiplexing, Queuing and Packet Loss



Multiplexing, Queuing at Output Link



Multiplexing, Queuing at Output Link

